MOBILE LANGUAGE LEARNING DESIGN ENVIRONMENT (MLLDE) · DESIGN CONTRACT

Curriculum Design Blueprint

A replicable framework for gamified, 5-minute mobile language lessons

v1.0 · Level A1–A2 · First lesson: “Language Passport” — Barcelona, De Tapas Por La Rambla

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This blueprint is the framework for the whole Curriculum. “Language Passport” — the Barcelona De Tapas Por La Rambla *travel-quest — is one lesson built from it, used throughout as the worked example.*

Status: v1.0 (design contract) · **Level:** A1–A2, millennial learners · **First build:** Barcelona — *De Tapas Por La Rambla*

This blueprint is the **replicable design contract** behind the Language Passport game. It is deliberately **destination- and language-agnostic**: the game is an *engine + data* system, and a new city or language is authored by filling the the Destination Schema (§7) — not by rewriting code. Everything here is grounded in the attached source matrix (*Educational Resources and Research on Digital Learning*); citations use `[S#]` keyed to Sources (§12).

1. Design goal

Deliver **standalone 5-minute mobile missions** that take an A1–A2 learner from passive exposure to **authentic communicative output**, wrapped in a travel narrative and a persistent gamified economy — replicable across destinations and languages, and shippable in two tiers (free/offline and Pro/AI).

Mobile learning demands its *own* design logic, distinct from e-learning: brevity, relevance, and usability for fragmented attention `[S3]`. 25% of users abandon an app after a single use `[S9]`, so every second of the 5-minute loop must earn the next.

2. Theoretical foundations (synthesised from the source matrix)

Framework	Contribution to the design	Source
SAM (Successive Approximation Model)	Agile build method: Preparation → Iterative Design (Savvy Start) → Iterative Development (Alpha/Beta/Gold). Rapid prototyping over linear planning.	[S-SAM]
G-CLIL (Gamified CLIL)	Four-ring concentric model: game mechanics *are* language tasks; disciplinary genre is the structural engine; narrative beats = identity formation.	[S-GCLIL]
CLIL 4Cs (Coyle)	Content · Communication · Cognition · Culture — every mission must touch all four.	[S-GCLIL]
IPO (Casañ-Pitarch)	Input → Processing → Output cycle: encounter, manipulate, deploy language.	[S-IPO]
ESA (Harmer)	Session-level sequencing: Engage → Study → Activate; variants Straight Arrow / Boomerang / Patchwork.	[S-ESA]
CTL (Johnson)	Brain-compatible learning via 8 components + authentic assessment.	[S-CTL8]
REACT	Relating · Experiencing · Applying · Cooperating · Transferring — CTL made actionable.	[S-REACT]
Mobile Learning Design	Context-driven design, microlearning, empathy-based design, Bloom's adaptation, "Using AI" prompts.	[S3]
GenAI competency (e-learning principles)	Multimedia, pre-training, learn-by-doing, scaffolding; equitable gains.	[S4]
Responsible Human–AI Collaboration	AI mentor scaffolds, never replaces, the learner; AI is a tutor, not an answer key.	[S-AWJ]
UI/UX best practices + HIG / Material	Progressive disclosure, low cognitive load, touch targets, meaningful motion.	[S9]
AWJ precedent	Narrative identity arc (Novice→Professor Prodigy) + AI mentor as a proven pattern.	[S-AWJ]

3. The integrated design model

The frameworks are not alternatives — they nest:

```
SAM  ── how we BUILD it (agile, iterative)
└ G-CLIL 4 Rings ── how the GAME is STRUCTURED
    └ Ring 1  Disciplinary genre & content   (the "why": real scenario)
      Ring 2  Pedagogical purpose            (CTL objectives, ESA phase)
      Ring 3  Game genre                      (role-play travel sim)
      Ring 4  Mechanics/dynamics/emotions     (coins, souvenirs, energy…)
       └ IPO × ESA ── how each 5-min SESSION FLOWS
           └ CTL + REACT ── how learning stays MEANINGFUL & AUTHENTIC
               └ Mobile UX + GenAI principles ── how it FEELS on a phone
```

Session sequencing — IPO mapped to ESA:

Loop phase	IPO	ESA	CTL / REACT
Arrive (narrative hook)	—	**Engage**	Relating
Vocabulary (animated images)	**Input**	Study	Experiencing
Drills (gap-fill, drag-match)	**Processing**	Study	Applying
NPC ordering mission	**Output**	**Activate**	Applying · Cooperating
Reflection (podcast/postcard)	**Output**	Activate	Transferring

4. Narrative & identity (G-CLIL Ring 3 + AWJ pattern)

- **Arc:** the learner rises **from Hitchhiker → Travel Ambassador**, echoing the proven AWJ identity transformation (Novice Writer → Professor Prodigy) [S-AWJ].

- **AI mentor:** a friendly travel-guide mentor scaffolds the learner (the Language Passport analogue of PhD Chema), giving **hints, never answers** — Responsible Human–AI Collaboration `[S-AWJ]`.
- **Narrative transportation:** learner **choices advance the plot and carry consequences** (ordering wrong drains energy) — they are a participant, not a spectator.
- Each destination is a **stamped page** in a growing story; the passport is the spine of the narrative.

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5. Vocabulary as animated dual-coded images (new core principle)

Grounded in the **multimedia principle** — people learn better from words + images than words alone — and consistent-cue retrieval `[S4]`, reinforced by mobile brevity/relevance `[S3]`:

1. **Every vocabulary item carries an image** (offline: an **inline SVG illustration**, e.g. *patatas bravas*, *tortilla*, *cervecilla*, *pinchos*).
2. On **Input**, each image plays a **defined entrance animation** (e.g. a themed "pop-in": scale + fade + slight rotate). The animation is a **named property of the vocab item**, not incidental styling.
3. When the term is later needed and the learner taps **"pista / hint"** during Processing or the NPC mission, the **same image replays the exact same animation** in a hint popover.
4. **Why:** identical encoding and retrieval cues strengthen the memory trace and give the learner a stable, recognisable anchor — the picture *is* the mnemonic.

Spec: `vocab[i] = { term, translation, svg, anim }` where `anim` is one of a small named set (`popIn`, `slideUp`, `flip`…). The Input renderer and the hint renderer call the **same** animation routine, guaranteeing they match.

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6. Objectives & authentic assessment (CTL)

Action-oriented objectives — state the knowledge with active verbs, and state the **benefit** so the learner sees personal meaning `[S-CTL8]`:

> *"By the end of this mission you will order food and drinks at a Spanish bar like a local — so you'll never freeze at a counter in Barcelona again."*

Authentic assessment, not multiple-choice `[S-CTL8]`: the learner is judged on **real in-game performance** — successfully getting the *camarero* to serve a full order **before the energy/hambre meter empties**. "Excellent achievement" = complete order + correct register (`nos pone…`) + no energy lost.

Each mission touches all **CLIL 4Cs**: Content (tapas domain), Communication (ordering), Cognition (choosing register/quantities), Culture (La Rambla facts).

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7. The replicable Destination Schema

This is the replicability engine. New destination/language = new data object; the engine is untouched.

```
Destination = {
  id, city, country, language, flag,          // "barcelona","Spain","es-ES","🇪🇸"
  level,                                       // "A1-A2"
  genre,                                       // disciplinary genre / scenario
  objective: { verbStatement, benefit, excellentAchievement },   // CTL
  fourCs: { content, communication, cognition, culture },        // CLIL

  narrative: { arrivalHook, identityFrom, identityTo, mentorName, mentorAvatar },

  vocab: [ { term, translation, svg, anim } ],                   // §5 animated images

  model: [ { speaker, text } ],               // Input dialogue (TTS)

  processing: {
    gapFill:   [ { sentence, answer } ],
    dragMatch: [ { term, target } ]
  },

  mission: {                                  // Output — authentic assessment
    npc, goal, energyStart,
    freeScript,                               // branching dialogue tree (offline)
    proPrompt,                                // LLM system prompt (Pro tier)
    successCriteria, hints
  },

  reflection: {                              // Output — choose your skill
    oral:    { prompts:[ "travel podcast", "call your mom" ], sttLang },
    written: { prompt: "postcard to a friend" },
    rubric                                    // targets: vocab used, min length…
  },

  culture: [ { icon, title, fact } ],         // "¿Sabías que…?"

  rewards: {
    stampSvg,
    souvenir: { name, svg, cost },            // e.g. Miró-mosaic tile
    nextFlight: { destinationId, cost }       // "buy a ticket to the next country"
  }
}
```

To add a new destination/language: author one `Destination` object (new vocab + SVGs, new scenario, new souvenir), set `nextFlight`, done. To add a new **language**, also set `language` / `sttLang` for TTS/STT.

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8. Gamified economy (G-CLIL Ring 4) — every mechanic traces to a learning aim

Mechanic	Learner meaning	Ties back to
Coins 🪙	Earned by completing Processing/Output correctly; **spent to buy a flight** to the next country.	Mastery, progression
Energy / hambre meter	Depletes on ordering errors in the NPC mission.	Authentic assessment `[S-CTL8]`
Passport stamp 🛂	Awarded per completed destination — proof of the journey.	Significant work, narrative
Souvenir 🧳	Bought with coins, stored in the **carry-on bag** (persistent inventory).	Collection, autonomy
Flight ticket ✈️	Unlocks the next destination.	Progression, goal-setting
Rank	Hitchhiker → … → Travel Ambassador.	Identity transformation `[S-AWJ]`

Persistence: passport, coins, and carry-on bag are saved in `localStorage` (per-browser, no login) so the meta-game carries across visits.

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9. Mobile UI/UX requirements [S9]

- **Progressive disclosure / onboarding:** teach controls *in* the first level, don't front-load instructions.
- **Low cognitive load:** one task on screen at a time; brevity and scanability.
- **Touch targets:** min ~44×44pt, ample spacing (iOS 375pt / Android 360dp reference widths).
- **Immediate feedback:** micro-interactions (color, motion) confirm every action instantly — meaningful motion, not decoration.
- **Empathy-based design:** built for 5-minute windows in distracting environments.

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10. Delivery tiers (Conflict-1 resolution — Hybrid)

Hosting	Self-contained HTML, GitHub-Pages-ready, no network	Same shell + network + API key
NPC mission	Scripted **branching** *camarero* (`freeScript`)	**LLM** conversation (`proPrompt`)
Reflection feedback	Rule-based rubric check	LLM feedback on podcast/postcard
Speech	Web Speech **TTS**; **STT** where supported	Same + AI transcription/feedback

The mock-up ships the **free** experience in full and exposes an **"Upgrade to Pro ✨"** path so the portfolio demonstrates both tiers.

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11. Build methodology (SAM)

1. **Preparation** — this blueprint + learner/mobile constraints (done).
2. **Iterative Design (Savvy Start)** — Barcelona vertical slice as the low-fi→hi-fi prototype.
3. **Iterative Development** — Alpha (engine + Barcelona data, offline), Beta (Pro AI hooks + polish), Gold (portfolio Before↔After index).

Deliverables: **BEFORE** (`examples/de-tapas-por-la-rambla.original.html` , already saved) ↔ **AFTER** (`Language Passport`), plus this blueprint.

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12. Sources

Keyed to the attached matrix *Educational Resources and Research on Digital Learning* and its "Referencias de fuentes" index.

Key	Resource (matrix)	Underlying reference
`[S-AWJ]`	The Academic Writing Journey (AWJ)	Final Tarea 3 Simulation.pdf `[1,2]`
`[S3]`	Clothier, P. (2026). *Mastering Mobile Learning Design: A Practical Guide* (1st ed.). Routledge.	BOOK REVIEW 1.docx `[3]`
`[S4]`	Asynchronous modules for genAI knowledge/self-efficacy	1-s2.0-S0360131526000783-main.pdf `[4,5]`
`[S-CTL8]`	Contextual Teaching and Learning — 8 components (Johnson)	Elaine Johnson, CTL `[6,7]`
`[S-REACT]`	CTL: Preparing students for the new economy (REACT)	`[6]`
`[S-GCLIL]`	G-CLIL (Casañ-Pitarch four rings + Coyle 4Cs)	Final Tarea 3 Simulation.pdf `[1,2]`
`[S-SAM]`	Successive Approximation Model (SAM)	`[1,2]`
`[S-IPO]`	Casañ-Pitarch's IPO Framework	`[1,2]`
`[S-ESA]`	Harmer's ESA Model	`[2]`
`[S9]`	UI/UX Best Practices + Apple HIG / Material Design	UI/UX guides `[9]`
`[S-DT]`	De Tapas Por La Rambla (v1.2)	Storyboard `[11]` , CLT Info-Gap `[12]`

Full bibliographic details live in the source workbook.